

Acute effects of conventional versus wide-pulse neuromuscular electrical stimulation on quadriceps evoked torque and neuromuscular function

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Funding information

Doctoral Research Grant, Jean Monnet of Saint-Etienne

Purpose: The effectiveness of a neuromuscular electrical stimulation (NMES) program is proportional to the level of evoked torque, which can be achieved with either conventional or wide-pulse stimulations. The aim of this study was to compare evoked torque, objective fatigability, and related peripheral and central alterations, as well as changes in central nervous system (CNS) excitability induced by an acute session of conventional versus wide-pulse NMES.

Methods: Seventeen young men underwent three 20-min NMES sessions: conventional (0.2 ms/50 Hz), wide-pulse at 50 Hz (1 ms/50 Hz), and wide-pulse at 100 Hz (1 ms/100 Hz). Neuromuscular measurements (i.e., maximal voluntary contraction, voluntary activation, evoked responses to femoral nerve stimulation, and CNS excitability) were performed on the right quadriceps femoris muscle before and after each NMES session. CNS excitability was measured using transcranial magnetic, thoracic, and transcutaneous spinal cord stimulations.

Results: The level of evoked torque was not significantly different between conventional and wide-pulse protocols applied at the maximal tolerable current intensity. All NMES protocols induced objective fatigability (~14% decrease in maximal voluntary contraction torque, $p < 0.001$) associated with peripheral (decrease in doublet torque and potentiated M-wave amplitude, $p = 0.002$ and $p < 0.001$, respectively) but not central (unchanged voluntary activation, $p = 0.79$) alterations. However, these acute changes did not differ between NMES protocols and none of the NMES protocols modified markers of CNS excitability.

Conclusion: These results may allow to conjecture that chronic effects and treatment effectiveness could be comparable between conventional and wide-pulse NMES.

KEYWORDS

corticospinal excitability, fatigability, motoneuronal excitability, NMES, spinal excitability

1 | INTRODUCTION

Neuromuscular electrical stimulation (NMES), which consists of evoking contractions by applying an electrical current—mainly over the muscle belly but also occasionally over the nerve trunk—via surface electrodes, is commonly used in rehabilitation for the recovery/preservation/improvement of muscle mass and function.¹ The conventional use of NMES in clinical practice, mainly targeting the quadriceps femoris, usually encompasses narrow pulses (0.2–0.5 ms) delivered at a frequency of 30–50 Hz.² Patients are encouraged to attain their maximal tolerable current intensity, as achieving the highest level of evoked torque is a key factor for optimizing the effectiveness of a NMES program.^{3–5} Indeed, Lai et al.³ reported a linear dose–response relationship between NMES effectiveness and evoked torque, with training-induced strength gains of 24% and 48% for evoked torque levels of 25% and 50% of maximal voluntary contraction (MVC), respectively. However, the ability to increase current intensity is limited by the associated discomfort⁶ and the maximal stimulator output.⁷ Thus, it is of clinical importance to appraise NMES protocols that can maximize evoked torque.

Current charge (i.e., the product of current intensity, pulse duration, and stimulation frequency) is a strong predictor of NMES-evoked torque.^{8,9} However, current intensity cannot be infinitely increased because of the associated discomfort.^{6,10} Similarly, as suggested by the sigmoid torque–frequency relationship, evoked knee extension torque increases with increasing stimulation frequency until 50–60 Hz, but then reaches a plateau.^{8,11} Therefore, increasing pulse duration could be considered as an alternative strategy to maximize evoked torque.^{8,12,13} The use of wide pulses (~1 ms) seems indeed to increase the level of evoked torque with respect to conventional narrow pulses. For example, Liebano et al.¹⁴ observed a trend for a 5.5% greater maximal evoked torque with 1-ms compared with 0.4-ms pulses. While this difference between the two pulse durations was not significant, likely because of a relatively small number of subjects (i.e., 19 young women), our team recently demonstrated a significantly greater maximal evoked torque for wide-pulse (45% MVC) compared to conventional NMES (40% MVC) on a large population,⁷ these results being observed when using both a conventional frequency (50 Hz) and a higher frequency (100 Hz). However, previous studies comparing narrow- and wide-pulse stimuli delivered to the quadriceps femoris muscle at the maximal tolerable current intensity measured evoked torque during a single NMES train.^{7,14} Indeed, comparisons of conventional versus wide-pulse NMES repeated trains were previously investigated for the triceps surae muscle at submaximal current

intensities.^{15,16} Thus, it remains to be determined whether (i) the use of wide-pulses at the maximal tolerable current intensity would lead to a greater evoked torque throughout a full NMES session consisting of multiple trains delivered to the quadriceps femoris muscle, and if so, (ii) greater objective fatigability (i.e., decrease in MVC torque induced by a NMES session) would be observed with respect to conventional NMES with narrow pulses. Objective fatigability is an important limiting factor of NMES, especially in clinical populations who may show a premature and exaggerated decrease in maximal voluntary force due to the particular motor unit recruitment.¹⁷ This NMES-induced objective fatigability has usually been found to be related to peripheral alterations (i.e., neuromuscular propagation and contractile failure).^{15,18}

Besides the possibility of a greater evoked torque throughout a full NMES session, wide-pulse NMES is well known for favoring the activation of central pathways at low current intensities.^{19,20} For instance, wider pulses are known to depolarize sensory axons more effectively—due to their longer strength–duration time constant and lower rheobase than motor axons²¹—therefore leading to a greater transmission of afferent volleys to the central nervous system (CNS). As a result, it has been demonstrated that afferent volleys induced by electrical stimuli can modulate CNS excitability,²² in the same way as when afferent volleys are generated mechanically.²³ For instance, it was reported that a single session of wide-pulse NMES increased motor evoked potential (MEP) amplitude in response to transcranial magnetic stimulation.^{24–26} However, these findings were mainly obtained with nerve trunk stimulation, low current intensities (i.e., 15% of maximal M-wave^{27–30}), and long-duration NMES sessions (i.e., up to 120 min^{31–34}), which differ from NMES clinical practice recommendations.² In addition, the effect of wide-pulse NMES on CNS excitability may be frequency-dependent. For example, Mang et al.³⁰ reported that a 100-Hz stimulation frequency was more effective than 10 and 50 Hz in increasing MEP amplitude. It remains unknown whether the use of wide-pulses could similarly impact CNS excitability when NMES is applied at the maximal tolerable current intensity over the muscle belly (i.e., as usually recommended for rehabilitation purposes.³⁵) Overall, the changes in cortical, spinal, and motoneuronal excitability after a single session of such NMES protocols remain to be determined,³⁶ by further considering the possibility that such alterations could be frequency-dependent.^{37–39}

The aims of the present study were therefore to compare the acute effects of a single session of conventional versus wide-pulse NMES at low and high frequency on (1) the level of evoked torque, (2) the amount and origin of NMES-induced objective fatigability, and (3) CNS excitability (i.e., cortical, spinal, and motoneuronal

excitability). We adhered to clinical practice recommendations by delivering NMES to the quadriceps femoris muscle belly at the maximal tolerable current intensity. We hypothesized that (1) a greater level of evoked torque would be generated by wide-pulse NMES protocols, (2) this higher torque would lead to greater objective fatigability as well as changes in CNS excitability compared to conventional NMES, and (3) the high frequency protocol would induce greater objective fatigability and changes in CNS excitability than low frequency NMES.

2 | MATERIALS AND METHODS

2.1 | Subjects

Twenty-seven healthy men were initially screened for eligibility. Subjects were excluded if their maximal tolerable current intensity exceeded the maximal stimulator output (100 mA) during a first visit dedicated to inclusion and familiarization to the study procedures. Seventeen subjects (age: 27 ± 6 years, mass: 69 ± 5 kg, stature: 177 ± 7 cm) were ultimately included in the study. They were recreationally active and reported no history of neurological or musculoskeletal impairment. None of them had previously engaged in systematic NMES training. Subjects were asked to refrain from strenuous and unaccustomed physical activity for 24 h before testing to minimize the risk of neuromuscular fatigue or muscle damage. The study protocol was approved by the local ethics committee and was in accordance with the latest update of the Helsinki Declaration (except for registration in a database). All subjects gave their written informed consent before participation.

2.2 | Experimental design

Subjects visited the laboratory on four separate occasions, including a familiarization visit followed by three experimental sessions. The first visit aimed to familiarize subjects with neuromuscular testing and with each NMES protocol (i.e., progressive increments of current intensity up to the maximal tolerable current intensity). As illustrated on Figure 1, the experimental sessions included neuromuscular testing (see *Neuromuscular testing*) that was conducted before (PRE) and immediately after (POST) standardized NMES protocols (see *NMES protocols*). The experimental sessions were performed in a random order and at the same time of the day with 5 to 10 days between sessions. All experimental sessions were conducted on the right quadriceps femoris muscle under isometric conditions. To investigate NMES-induced objective fatigability, changes in MVC torque were quantified,

and the underlying central (i.e., voluntary activation (VA) measured using the interpolated twitch technique), and peripheral (i.e., neuromuscular propagation failure estimated with M-wave and contractile failure estimated with evoked doublet) alterations were further investigated. To determine NMES-induced changes in CNS excitability, we investigated MEPs with transcranial magnetic stimulation (i.e., corticospinal excitability⁴⁰), thoracic motor evoked potentials (TMEPs) with thoracic electrical stimulation (i.e., motoneuron excitability⁴¹), and spinal reflexes (SRs) with transcutaneous spinal cord stimulation (i.e., excitability of monosynaptic responses between Ia afferents and motor neurons⁴²).

2.2.1 | Neuromuscular testing

Testing began with the determination of femoral nerve stimulation intensity. Then, after a standardized warm-up (i.e., ten 3–5 s submaximal isometric contractions of progressively increasing intensity until reaching 90% of estimated MVC), baseline torque generating capacity was assessed. One MVC without stimulation followed by two MVCs with superimposed (i.e., 100-Hz paired stimuli delivered at maximal torque) and post-MVC femoral nerve stimulation (i.e., 100-Hz paired stimuli ~2 s following the MVC and single twitch separated by 3 s) in potentiated relaxed muscles were performed. The 100-Hz paired stimuli allowed estimation of VA using the interpolated twitch technique whereas the single stimulation allowed the associated EMG response to be obtained (potentiated M-wave; M_{POT}). During MVCs, subjects were instructed to contract their muscles as strongly as possible for ~4 s with a progressive torque buildup. A 1-min resting period separated the MVCs. Afterwards, the EMG associated with 20% MVC (20% MVC-EMG) was determined by performing a 5-s contraction at 20% MVC using visual feedback displayed on a computer monitor. Series of contractions at the target EMG were then performed to set the stimuli at appropriate intensities (see below). Baseline CNS excitability was then assessed with five sets of measurements each including four stimuli delivered in a random order every ~5 s (i.e., to record MEP, TMEP, SR, and M-wave responses; see Figure 1). Each set lasted ~20 s, and was separated by rest periods of 20 s. The NMES protocol started 1 min following the last set.

As NMES-induced CNS excitability changes can begin to recover within seconds after exercise cessation,⁴³ these assessments were distributed between the last six NMES-evoked contractions to avoid any recovery effect at POST (Figure 1B). During the 30-s resting period of NMES protocols, a set of stimuli was delivered during contractions targeting 20% MVC-EMG as determined at PRE.

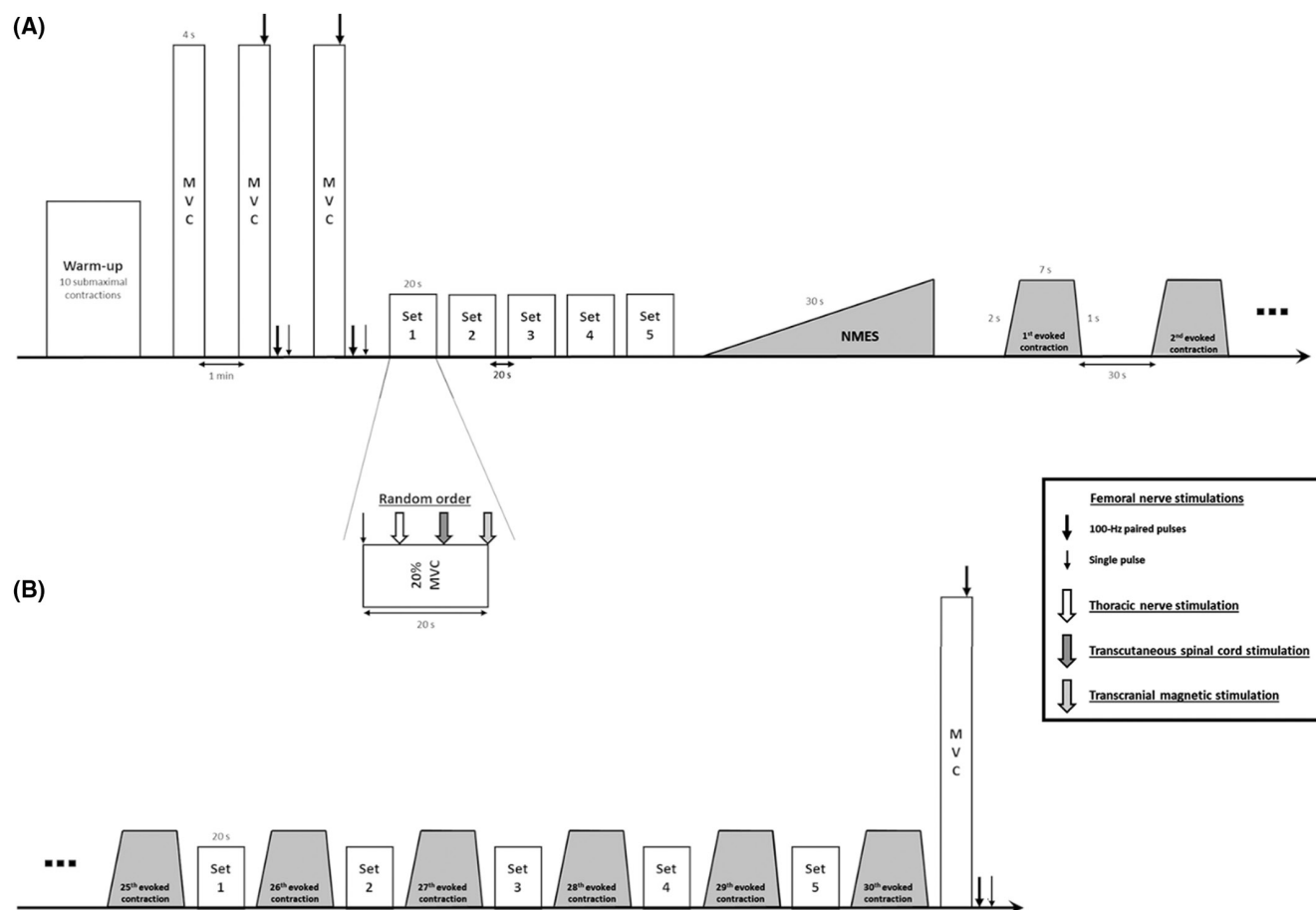


FIGURE 1 Experimental session design: neuromuscular testing (in white) at PRE (panel A) and POST (panel B), and NMES protocol (in gray). MVC, maximal voluntary contraction; NMES, neuromuscular electrical stimulation.

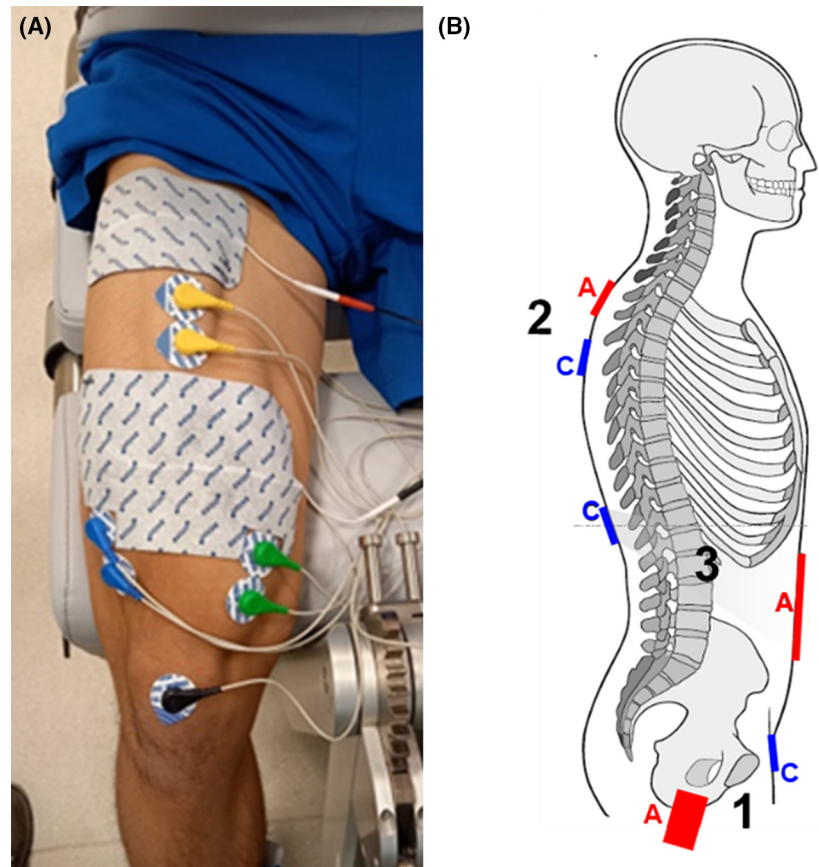
Immediately after the NMES protocols, torque generating capacity was assessed using one MVC with femoral nerve stimulation delivered on the MVC plateau and ~2 and 5 s after MVC in potentiated relaxed muscles.

2.2.2 | NMES protocols

Three NMES protocols were performed in a random order: narrow-pulse (0.2 ms) NMES applied at 50 Hz (CONV), wide-pulse (1 ms) NMES applied at 50 Hz (WP50), and wide-pulse (1 ms) NMES applied at 100 Hz (WP100). The current was delivered with a commercially available device (BioStim 2.1, Mazet Santé, Electronique du Mazet, Le Mazet Saint Voy, France), which provides biphasic, symmetrical, and rectangular pulses. Two 2-mm-thick self-adhesive electrodes were placed over the right quadriceps femoris muscle (Figure 2A). One electrode, measuring 198 cm² (180 × 110 mm, axion GmbH, Leonberg, Germany), was placed on the distal third of thigh length over the

vastus lateralis (VL) and vastus medialis (VM) muscles. A small part of the electrode was trimmed to allow EMG electrodes placement. The other electrode, measuring 85 cm² (125 × 86 mm, axion GmbH), was placed 5–7 cm below the inguinal ligament. Each 10-s stimulation train included a ramp up of 2 s, a steady tetanic phase of 7 s and a ramp down of 1 s. Each train was followed by a 30-s rest period, and a total of 30 NMES trains were delivered during each protocol (Figure 1). Current intensity was gradually increased by the subjects to their maximally-tolerated level during a test train of 30 s that was delivered at the beginning of the NMES protocol. Then, subjects were allowed to consistently adjust the current intensity to their maximum tolerable level until the 25th train of each protocol. The maximal tolerated current intensity varied between 27 and 100 mA. Current intensity remained constant from the 25th train to the end of the NMES protocol to maintain a stable setup for CNS excitability assessments. For each NMES protocol, current intensity and the associated discomfort were assessed after each block of five trains.

FIGURE 2 Illustration of electrodes placement. Location of EMG and NMES electrodes (panel A). Location of anode (A, red) and cathode (C, blue) electrodes for femoral nerve stimulation (1), thoracic stimulation (2), and transcutaneous spinal cord stimulation (3) (panel B).



2.3 | Instrumentation

2.3.1 | Torque and EMG recordings

Subjects were seated in an isometric dynamometer (ARS dynamometry; SP2 Ltd., Ljubljana, Slovenia) with the hips and the knees at 90° and 60° of flexion (with 0° referring to the extended neutral position), respectively. The leg was attached to the chair by a noncompliant strap just proximal to intermalleolar axis. Trunk and pelvis were also securely strapped to the chair to prevent compensatory movements during contractions. The knee extension torque signal was low-pass filtered (20 Hz) and analog-to-digital converted at a sampling rate of 2 kHz by a PowerLab system (16/35; ADInstruments, Bella Vista, Australia). EMG activity from the right VL, VM, and rectus femoris (RF) muscles was recorded with pairs of self-adhesive surface electrodes (Melitrace 100; Covidien, Mansfield, MA) in bipolar configuration, with electrodes placed 30 mm apart. The reference electrode was placed on the ipsilateral patella. Before electrode placement (Figure 2A), the skin was shaved, abraded, and cleaned with isopropyl alcohol to limit impedance. EMG signals were amplified with an octal bio-amplifier (ML138; ADInstruments), band-pass filtered (5–500 Hz), and analog-to-digital converted at a sampling rate of 2 kHz by the PowerLab system.

2.3.2 | Femoral nerve stimulation

Single and paired rectangular electrical pulses of 1 ms (400-V maximal output voltage) were delivered via a constant-current stimulator (DS7R; Digitimer, Hertfordshire, UK) to the right femoral nerve. A 30-mm diameter surface cathode (Meditrace 100) and a 50×100 mm anode (Durastick Plus; DJO Global, Vista, USA) were taped to the skin onto the femoral triangle and in the gluteal fold (Figure 2B). Electrical stimuli were first delivered at 50 mA and then the intensity was increased with 10-mA increments until knee extension peak twitch torque and maximal M-wave responses were observed. The resulting stimulation intensity was then further increased by 30% (supramaximal intensities: 116 ± 46 mA, 127 ± 42 mA, and 121 ± 52 mA for CONV, WP50, and WP100 sessions, respectively; $p=0.22$) and maintained throughout the sessions for the interpolated twitch technique and for evoking the M-wave during the 20% MVC-EMG contractions (M_{SUP}), that was used to normalize all the other evoked EMG responses.

2.3.3 | Thoracic stimulation

A constant-current stimulator (DS7R) was used to evoke TMEP by delivering a single rectangular electrical pulse

of 1 ms between surface electrodes (Meditrace 100) positioned between the spinous processes of T1 and T2 (anode) and between T3 and T4 (cathode) (Figure 2B). The stimulation intensity was adjusted to elicit a VL TMEP amplitude of 20% VL M_{SUP} during an ongoing 20% MVC-EMG contraction. Mean stimulation intensities were 452 ± 191 mA, 473 ± 195 mA, and 471 ± 209 for CONV, WP50, and WP100 sessions, respectively ($p = 0.43$).

2.3.4 | Transcutaneous spinal cord stimulation

Subjects were asked to maintain a symmetrical position of the upper limbs and their head as immobile as possible in a neutral position. Single rectangular electrical pulses of 1 ms were delivered via a constant-current stimulator (DS7R). A 68×125 mm anode (axion GmbH) was placed over the midline of the abdomen between the xiphoid process of the sternum and the umbilicus, and a 50×50 mm cathode (Durastick Plus) was placed on the midline of the back between the spinous processes of L1 and L2 (Figure 2B). To confirm that the responses were caused by activation of the sensory fibers, a rectangular double-pulse stimulation (interpulse interval: 50 ms) was delivered to test homosynaptic depression of the evoked responses.⁴² This phenomenon was confirmed in the present study as the responses associated to the second transcutaneous spinal cord stimulation were noticeably smaller than those following the first stimulus for all NMES protocols (4.0 ± 1.6 and 0.7 ± 0.6 mV, respectively). Stimulus intensity was adjusted to evoke VL SR with an amplitude of 20% of VL M_{SUP} during an ongoing 20% MVC-EMG contraction. Mean stimulation intensities were 115 ± 33 , 116 ± 29 , and 120 ± 30 mA for CONV, WP50, and WP100 sessions, respectively ($p = 0.43$).

2.3.5 | Transcranial magnetic stimulation

Single-pulse transcranial magnetic stimulation was delivered to the left motor cortex via a concave double cone coil using a magnetic stimulator (Magstim 200²; The Magstim Company Ltd.). The optimal coil location was established for each visit as the position that elicited the largest MEP in the VL muscle during a low-level voluntary contraction (20% MVC). This position was marked on a swim cap that was secured to the scalp. Subsequently, transcranial magnetic stimulation intensity was adjusted to evoke VL MEP with an amplitude

of 20% of VL M_{SUP} during an ongoing 20% MVC-EMG contraction. Mean stimulation intensities were 48 ± 13 , 49 ± 14 , and $48 \pm 12\%$ of maximum stimulator output for CONV, WP50, and WP100 sessions, respectively ($p = 0.73$).

2.3.6 | Discomfort assessment

Subjects were asked to place a vertical mark on a horizontal line (100 mm) to rate the level of discomfort. The scale ranged from 0 to 100 mm, with the 0-mm value representing “no discomfort” and the 100-mm value representing “worst possible discomfort”. The mean discomfort was calculated for each NMES protocol by averaging the five discomfort assessments.

2.4 | Data analysis

2.4.1 | Voluntary and evoked torque

Objective fatigability was quantified as the PRE-to-POST decline in MVC torque induced by each NMES session. MVC torque was determined as the highest voluntary torque recorded during the three MVCs. The amplitude of the doublets evoked by 100-Hz paired pulses (Db100) was measured and VA was calculated as follows:

$$VA = 1 - \left[\text{superimposed Db100} \times (\text{torque at stimulation} / \text{maximal torque before stimulation}) / \text{Db100} \right] \times 100.$$

Db100 and VA were measured from the trial where the MVC torque was the highest before the superimposed paired stimuli were delivered.⁴⁴ Alterations associated to objective fatigability were considered as PRE-to-POST decreases of Db100 and M_{POT} (i.e., peripheral alterations) and VA (i.e., central alteration).

The torque-time integral (TTI) of each NMES-evoked contraction was quantified for the entire 10-s stimulation period and evoked peak torque (EPT) for each NMES-induced contraction was defined as the highest torque reached during each train and was expressed as a percentage of MVC torque (Figure 3A). A first analysis of TTI and EPT was focused on the first NMES-evoked contraction. A second analysis of these parameters was conducted for blocks of five NMES-evoked contractions (block 1: contraction 1–5, block 2: 6–10, block 3: 11–15, block 4: 16–20, block 5: 21–25, and block 6: 26–30). Finally, a global analysis was done on the total TTI (i.e., sum of TTI of the

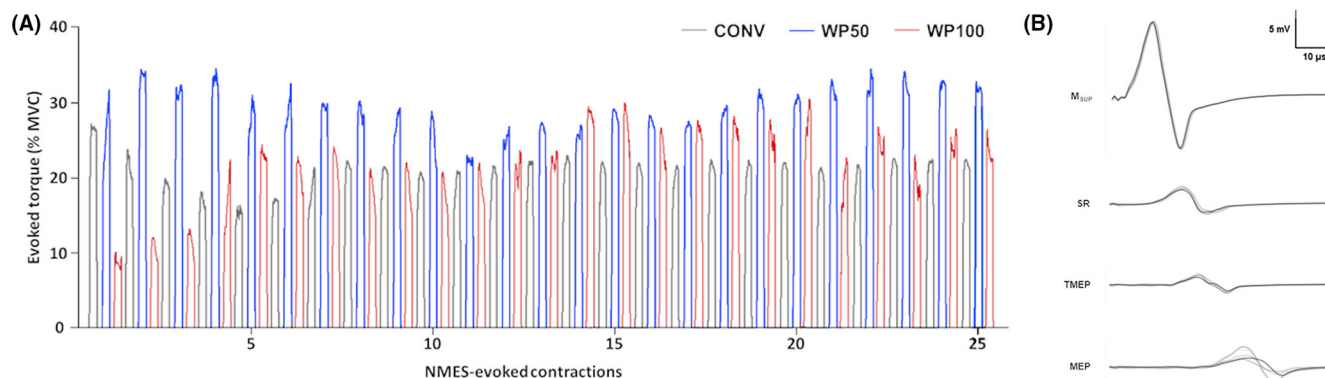


FIGURE 3 Example for a representative subject of evoked torque traces for each NMES protocol narrow-pulse 50 Hz (CONV), wide-pulse 50 Hz (WP50) and wide-pulse 100 Hz (WP100) (panel A) and EMG traces from vastus lateralis in response to femoral nerve (M-wave evoked during submaximal contractions, M_{SUP}), transcutaneous spinal cord (spinal reflex, SR), thoracic (thoracic motor evoked potential, TMEP) and transcranial magnetic (motor evoked potential, MEP) stimulations (overlays of five sets of measurements) (panel B).

30 NMES-evoked contractions) and the mean EPT (i.e., mean EPT of the 30 NMES-evoked contractions).

2.4.2 | EMG

The 20% MVC-EMG was determined by calculating the average root mean square EMG of VL from the 20% MVC. To assist subjects in visualizing and maintaining the ongoing 20% MVC-EMG contraction, the VL root mean square EMG was smoothed with a 400-ms time window. For all tested muscles, M_{POT} , M_{SUP} , SR, TMEP, and MEP peak-to-peak amplitudes were also measured and then averaged from the five sets (Figure 3B). SR, TMEP, and MEP were consistently expressed as a percentage of M_{SUP} . Comparing changes in MEP and TMEP (MEP/TMEP ratio) allowed us to distinguish neural changes between supraspinal and spinal levels.³⁶ As supported by recent findings, SR and H reflex are two neurophysiological responses involving the same spinal pathway.⁴⁵ SR amplitude can be therefore influenced by motoneuron excitability as well as by the efficacy of synaptic transmission (i.e., presynaptic inhibition and homosynaptic postactivation depression).^{42,46} As stimulation of the descending tracts directly activates large-diameter axons (not subjected to presynaptic inhibition), TMEP amplitude may be used as a proxy of motoneuron excitability.⁴⁷ Therefore, comparing SR and TMEP (SR/TMEP ratio) should allow us to distinguish whether potential changes in spinal excitability were presynaptic (i.e., efficacy of Ia-afferent— α -motoneuron pathway) or postsynaptic (i.e., motoneuron excitability) in origin.

2.5 | Statistical analysis

Statistical analyses were performed with JAMOVI software (version 1.1.9; <https://www.jamovi.org/download.html>). Shapiro–Wilk and Levene tests were used to examine data normality and homogeneity of variances, respectively. All data are presented as mean values $\pm$ SD. Statistical significance was set at $p < 0.05$.

One-way repeated measure ANOVAs were performed to compare the intensities of each stimulation technique (i.e., femoral nerve, transcutaneous spinal cord, thoracic, and transcranial magnetic stimulation) between NMES protocols. One-way repeated measure ANOVAs were also used to evaluate differences between NMES protocols at PRE for all neuromuscular variables (i.e., MVC torque, Db100, VA, M_{POT} , SR, TMEP, MEP, SR/TMEP, and MEP/TMEP). One-way repeated measure ANOVAs were conducted to compare the TTI and EPT of the first NMES-evoked contraction, and the total TTI and mean EPT of the whole NMES session between NMES protocols. As discomfort data were not normally distributed, a Friedman test was used to compare mean discomfort levels between NMES protocols. Two-way repeated measure ANOVAs (block of 5 NMES-evoked contractions [1 vs. 2 vs. 3 vs. 4 vs. 5 vs. 6] $\times$ NMES protocol [CONV vs. WP50 vs. WP100]) were performed to analyze the evolution of current intensity, TTI and EPT during and between NMES protocols. Two-way repeated measure ANOVAs (time [PRE vs. POST] $\times$ NMES protocol [CONV vs. WP50 vs. WP100]) were also used for all neuromuscular variables. Tukey post-hoc tests were performed when the ANOVAs revealed a significant effect. Effect size is only presented for significant findings as partial eta squared

(η_p^2) with values representing small ($\eta_p^2 \geq 0.01$), medium ($\eta_p^2 \geq 0.06$), and large ($\eta_p^2 \geq 0.14$) effects.⁴⁸

3 | RESULTS

3.1 | Neuromuscular electrical stimulation protocols

Current intensity changed throughout NMES session ($p < 0.001$; $\eta_p^2 = 0.78$), as it increased progressively during the first four blocks of contractions ($p \leq 0.03$) and then plateaued after block 4 ($p = 0.22$; Figure 4A). Current intensity was different between NMES protocols ($p < 0.001$; $\eta_p^2 = 0.93$), being higher for CONV compared with WP50 and WP100 ($p < 0.001$), with no difference between WP50 and WP100 ($p = 0.58$; Figure 4A). For the first NMES-evoked contraction, TTI and EPT were similar between NMES protocols (TTI: 546 ± 260 Nm.s for CONV, 614 ± 419 Nm.s for WP50, and 462 ± 294 Nm.s for WP100; EPT: $26.8 \pm 14.1\%$ MVC for CONV, $32.0 \pm 17.6\%$ MVC for WP50, and $25.9 \pm 13.7\%$ MVC for WP100; $p = 0.11$ and $p = 0.23$, respectively). For the whole NMES session, TTI and EPT were similar between NMES protocols ($p = 0.19$ and $p = 0.23$, respectively). Yet, TTI and EPT changed during NMES session ($p < 0.001$; $\eta_p^2 \geq 0.34$). Block 6 showed lower TTI than the other blocks ($p < 0.001$), except block 1 ($p = 0.12$). Block 1 showed lower EPT compared with blocks 3 and 4 ($p = 0.009$), and EPT of block 6 was also lower than the other blocks ($p < 0.001$), excepted block 1 ($p = 0.52$) (Figure 4B,C). Total TTI, mean EPT, as well as mean discomfort (CONV: 64.8 ± 28.1 mm; WP50: 66.3 ± 28.6 mm; WP100: 70.8 ± 26.8 mm) were similar between NMES protocols (all $p \geq 0.17$; Figure 5).

3.2 | Objective fatigability

MVC torque, Db100, VL M_{POT} , and VA data are presented in Table 1. At PRE, MVC torque, Db100, VL M_{POT} , and VA were similar between NMES protocols (all $p \geq 0.07$). MVC torque, Db100, and VL M_{POT} were reduced from PRE-to-POST (all $p \leq 0.002$; $\eta_p^2 \geq 0.46$; Figure 6A). VA was not changed after the NMES session regardless of the condition ($p = 0.79$; Figure 6A).

3.3 | Central nervous system excitability measurements

We only present the evoked EMG responses for the VL muscle, as we set stimulation intensities targeting evoked

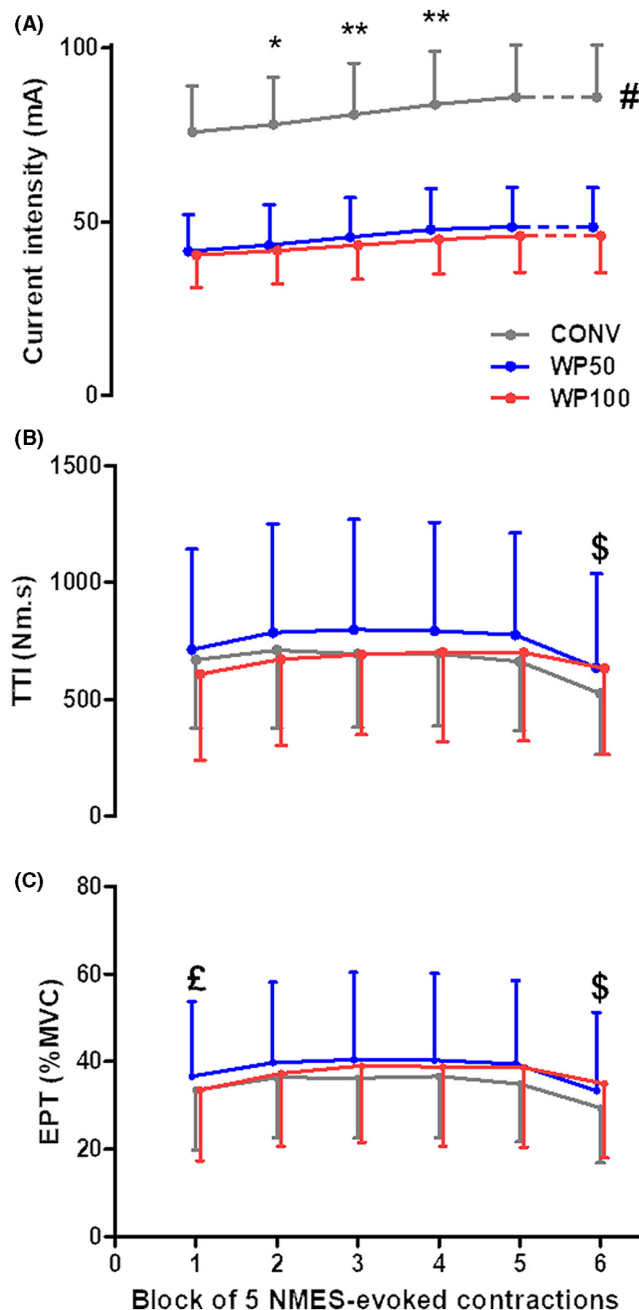


FIGURE 4 Evolution of current intensity (panel A), torque-time integral (TTI; panel B) and evoked peak torque (EPT, panel C) during narrow-pulse 50 Hz (CONV, gray lines), wide-pulse 50 Hz (WP50, blue lines) and wide-pulse 100 Hz (WP100, red lines) NMES protocols. CONV higher than WP50 and WP100: # $p < 0.001$. Block higher than the preceding block: * $p < 0.05$ and ** $p < 0.01$. Block lower than the 2nd, 3rd, 4th and 5th blocks: \$ $p < 0.001$. Blocks lower than the 3rd and 4th blocks: £ $p < 0.01$. MVC, maximal voluntary contraction.

responses of 20% of VL M_{SUP} during a contraction performed at 20% MVC-EMG determined on the VL. Yet, as can be seen in Appendix S1 (<https://figshare.com/s/c84a6b315575e81364b3>), VM and RF behaved similarly. TMEPs were obtained from 16 subjects due to excessive discomfort for one subject.

Data on CNS excitability are presented in Table 2. At PRE, SR, TMEP, MEP, SR/TMEP, and MEP/TMEP were similar between NMES protocols (all $p \geq 0.45$). All evoked EMG responses were unchanged after the NMES session regardless of the condition (all $p \geq 0.20$, Figure 6B).

4 | DISCUSSION

The purpose of the present study was to compare (1) evoked torque, (2) objective fatigability, and related peripheral and central alterations, as well as (3) changes in CNS excitability, induced by an acute session of conventional versus wide-pulse NMES. Contrary to our hypothesis, the level of evoked torque was not significantly different between the NMES protocols. NMES-induced peripheral, but not central alterations, that did not differ between protocols, that is, objective fatigability was similar for conventional and wide-pulse NMES. None of the NMES protocols modified markers of CNS excitability.

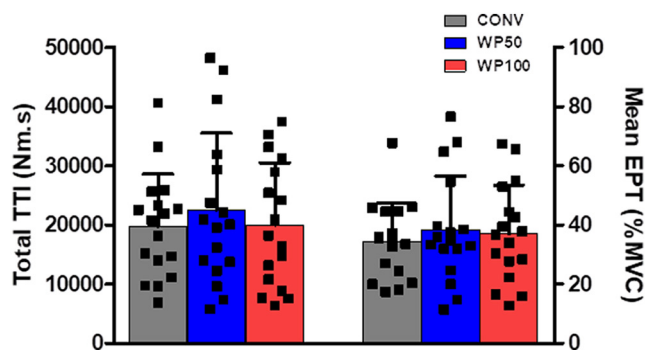


FIGURE 5 Total torque-time integral (TTI) and mean evoked peak torque (EPT) for narrow-pulse 50 Hz (CONV), wide-pulse 50 Hz (WP50), and wide-pulse 100 Hz (WP100) NMES protocols. Columns, error bars and squares indicate mean, SD, and individual data, respectively.

4.1 | Neuromuscular electrical stimulation session

Contrary to our hypothesis, we did not observe a significant difference in the level of evoked torque between conventional and wide-pulse protocols (i.e., approximately 27% vs. 29% of MVC) when considering only the first NMES-evoked contraction. This result is in accordance with the findings of Liebano et al.¹⁴ who failed to demonstrate a significantly greater maximal evoked torque with 1-ms compared with 0.4-ms pulse duration trains in 19 subjects, yet they interpreted these results as a trend that could lead to significant differences with additional subjects. However, we recently reported a significantly greater maximal evoked torque for WP50 and WP100 than for CONV NMES (i.e., approximately 45% vs. 40% of MVC) for a single NMES train in 50 subjects⁷ that partially supports the hypothesis that wide pulses may lead to greater evoked torque than narrow pulses. Indeed, there is a high intersubject variability for maximal evoked torque (see Figure 1 of Espeit et al.⁷), which may mask some differences between conditions that could only arise when considering a large population. Moreover, the weak level of evoked torque measured for the first NMES-evoked contraction in the present study (<30% of MVC) is in accordance with previous observations.^{49,50} This may reflect a certain reluctance of subjects to increase current intensity at the beginning of a NMES session and the necessity of an accommodation to NMES before initiating a training program.

When the entire NMES sessions was considered in the present study, we did not observe any difference between conventional and wide-pulse protocols for the mean level of evoked torque (i.e., ~37% MVC), in agreement with the 25%–45% MVC range classically observed in the literature.^{18,49–52} However, subjects required higher intensities to achieve the maximal tolerable level with conventional than wide-pulse NMES protocols, in

TABLE 1 Neuromuscular parameters before (PRE) and after (POST) each NMES protocol.

	CONV		WP50		WP100	
	PRE	POST	PRE	POST	PRE	POST
MVC torque (Nm)	272 ± 47	232 ± 49	278 ± 48	235 ± 49	276 ± 51	242 ± 50
Db100 (Nm)	105 ± 28	97 ± 25	107 ± 31	100 ± 30	108 ± 29	102 ± 27
M _{POT} (mV)	17.4 ± 4.2	14.7 ± 4.7	16.7 ± 4.3	14.4 ± 4.6	17.6 ± 4.5	14.0 ± 4.8
VA (%)	85.1 ± 6.2	85.7 ± 8.8	86.2 ± 5.9	85.6 ± 8.5	86.5 ± 5.2	87.6 ± 7.2

Note: Values are means ± SD. PRE and POST, before and after each NMES protocol.

Abbreviations: CONV, narrow-pulse at 50 Hz; Db100, 100-Hz doublet; M_{POT}, potentiated M-wave; MVC, maximal voluntary contraction; VA, voluntary activation; WP100, wide-pulse at 100 Hz. WP50, wide-pulse at 50 Hz.

* and ** lower than PRE ($p < 0.01$ and $p < 0.001$, respectively).

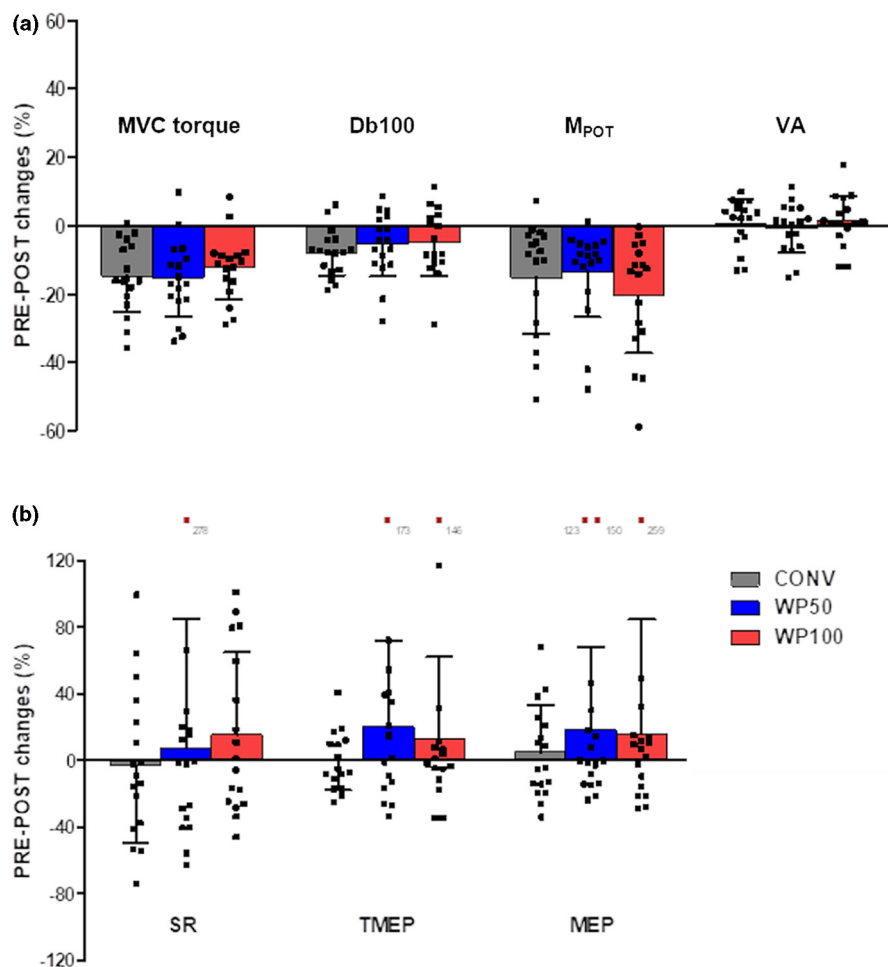


FIGURE 6 PRE-to-POST changes in objective fatigability (maximal voluntary contraction (MVC) torque, 100-Hz doublet torque (Db100), vastus lateralis potentiated M-wave (M_{POT}), and voluntary activation (VA); panel A) and in central nervous system excitability measurements (vastus lateralis spinal reflex (SR), thoracic motor evoked potential (TMEP), and motor evoked potential (MEP); panel B) induced by each NMES protocol (CONV, WP50, and WP100). Columns, error bars and squares indicate mean, SD, and individual data, respectively. Individual values of subjects with out-of-scale data are presented as red squares.

TABLE 2 CNS excitability before (PRE) and after (POST) each NMES protocol.

	CONV		WP50		WP100	
	PRE	POST	PRE	POST	PRE	POST
SR (% M _{SUP})	20.2 ± 6.3	19.7 ± 11.5	18.5 ± 6.8	17.1 ± 6.9	18.9 ± 6.3	20.8 ± 8.7
TMEP (% M _{SUP})	16.2 ± 5.4	15.6 ± 4.7	15.3 ± 4.7	17.5 ± 7.0	16.3 ± 5.0	17.7 ± 7.0
MEP (% M _{SUP})	21.1 ± 6.4	21.4 ± 7.7	21.6 ± 7.2	24.6 ± 12.4	22.7 ± 5.3	25.3 ± 11.1
SR/TMEP	1.44 ± 0.68	1.45 ± 0.90	1.38 ± 0.69	1.08 ± 0.42	1.29 ± 0.59	1.34 ± 0.98
MEP/TMEP	1.38 ± 0.49	1.42 ± 0.39	1.47 ± 0.58	1.41 ± 0.44	1.45 ± 0.48	1.41 ± 0.37

Note: Values are means ± SD. PRE and POST, before and after each NMES protocol.

Abbreviations: CONV, narrow-pulse at 50 Hz; MEP, motor evoked potential; SR, spinal reflex; TMEP, thoracic motor evoked potential; WP100, wide-pulse at 100 Hz. WP50, wide-pulse at 50 Hz.

line with previous reports in healthy young⁷ and old⁵³ adults. As the evoked torque is proportional to current charge (i.e., the product of current intensity and pulse duration), lower current intensities are usually needed with longer pulse durations.⁸ The lower current intensity with wide-pulse NMES reduces the likelihood of reaching the maximal stimulator output, leading to a smaller proportion of subjects limited by this practical issue.⁷

4.2 | Objective fatigability

The three NMES protocols induced similar levels of objective fatigability. These results corroborate previous findings reporting similar losses in MVC torque after conventional versus wide-pulse NMES at low and high frequency delivered at submaximal (and constant) current intensity to the triceps surae muscles^{15,54,55} or to the tibial nerve.⁵⁶ In the same way, similar MVC

torque losses have been reported after wide-pulse low frequency versus wide-pulse high frequency NMES sessions applied to the triceps surae muscles at an intensity evoking 30% of MVC⁵⁵ and to the tibial nerve at an intensity evoking 10% of MVC.⁵⁶ Yet, the observed reduction of ~14% MVC was lower than that usually reported (−20% to −30%) after a conventional NMES session at maximal tolerable current intensity on the quadriceps femoris.^{18,49,51,52} An explanation for this lower MVC torque decrease could be the different knee angle used here (60° of flexion) versus in the aforementioned studies (~90°). Foure et al.⁵⁰ demonstrated in fact that NMES at short muscle length (50° of knee flexion) induced less objective fatigability and muscle damage compared with long muscle length (100° of knee flexion). Objective fatigability could mainly be ascribed to the occurrence of peripheral alterations, as evidenced by decreases in Db100 and M_{POT}, with no or little implication of central factors (i.e., unchanged VA). Neuromuscular propagation failure (M_{POT} reduction of ~16%) and contractile failure (Db100 reduction of ~6%) have already been commonly reported following quadriceps^{18,51} and triceps surae NMES,^{15,55,57} despite contrary isolated results showing unchanged M-wave and potentiated peak twitch after low frequency NMES of the triceps surae.⁵⁸ Peripheral alterations could be explained by the repeated activation of the same muscle fibers arising from the spatially fixed motor unit recruitment imposed by NMES.⁵⁹ The unchanged voluntary activation has already been observed after conventional maximal-intensity NMES of the quadriceps^{18,51} and wide-pulse low-intensity NMES applied to the triceps surae muscle belly,¹⁵ probably as a consequence of the little CNS involvement. Again, conflicting results (voluntary activation failure) have been observed after low-⁵⁸ and high-⁵⁷ frequency NMES of the triceps surae. However, a possible effect of frequency on the origin of objective fatigability was discarded by Papaioordanidou et al.,⁵⁵ who reported similar objective fatigability associated with unchanged voluntary activation and decreased doublet torque following four NMES protocols entailing narrow- or wide-pulses at low or high frequency. Altogether, the etiology and the amount of objective fatigability was similar between NMES protocols, with peripheral but no central alterations.

4.3 | Central nervous system excitability measurements

Contrary to our hypothesis, we did not observe any significant changes in corticospinal excitability (unchanged MEP) for all NMES protocols applied at maximal tolerable current intensity. Despite some evidence suggesting

that conventional NMES may increase MEP^{60,61} and pulse duration may not influence changes in corticospinal excitability,²⁷ the lack of changes for MEPs was even more unexpected after wide-pulse NMES. This was particularly true for wide-pulse NMES at 100 Hz, as this frequency has been demonstrated to be the most effective for increasing corticospinal excitability.³⁰ Indeed, increases in corticospinal excitability were largely reported for the tibialis anterior after a session of wide-pulse NMES applied to the common peroneal nerve at submaximal current intensities^{24,26} and also for hand muscles following radial/ulnar nerve stimulation at weak current intensities,^{31,33} despite conflicting findings.⁶² We speculate that the discrepancies between the present and previous studies are due to experimental procedures. For instance, the application of wide-pulse NMES to the muscle belly rather than to the nerve trunk is known to minimize the involvement of the CNS.^{63,64} Moreover, to the best of our knowledge, no changes in MEP have ever been reported after a NMES session applied at the maximal tolerable current intensity, as submaximal intensities were used in previous studies (e.g., current intensity evoking 15% of M-wave or a visible contraction).^{30,33} In a systematic review, Chipchase et al.²² suggested that current intensity appeared to have the strongest relation with the modulation of corticospinal excitability, as 65% of experiments using current above motor threshold reported increased MEP whereas 42% of studies reported decreased MEP after stimulations below motor threshold. Furthermore, previous studies focusing on the time course of MEP following NMES of the tibialis anterior and first dorsal interosseus muscles, reported an increase that began to be significant after 24 and 45 min of wide-pulse NMES with a duty cycle of 50%, respectively.^{30,32} All the studies using wide-pulse NMES sessions shorter than 30 min failed to observe changes in MEP.^{38,65,66} Based on these considerations, it seems that applying NMES to the quadriceps muscle following clinical practice guidelines to improve/restore muscle function⁶⁷—as done here—may not influence corticospinal excitability. The use of longer NMES sessions, higher duty cycles and submaximal current intensities, as usually done in stroke rehabilitation,^{68,69} should probably induce cortical plasticity. Yet, NMES has been shown to favor sensorimotor activation, as revealed by functional magnetic resonance imaging investigations considering 5- and 13-min NMES protocols applied to the quadriceps⁷⁰ or triceps surae⁷¹ muscles.

Because MEP depends on the synaptic relays in corticospinal projections at the cortical level as well as at the spinal level,⁷² MEP should also be interpreted in light of changes at the spinal level.³⁶ Therefore, the present neuromuscular testing procedures allowed to investigate more accurately the potential changes at the cortical,

spinal, and motoneuronal levels. In addition to reporting unchanged corticospinal excitability, we did not observe any change of cortical, spinal, and motoneuronal excitability after conventional and wide-pulse NMES. At the spinal level, unchanged H-reflex has been found after conventional NMES applied over the triceps surae muscle belly⁵⁷ or to the tibial nerve.⁷³ Conflicting results have been observed after wide-pulse NMES at 20 Hz to 100 Hz, with no changes or decreases in H-reflex amplitude.^{73,74} Interestingly, Dideriksen et al.³⁷ found that conventional NMES applied at 100 Hz on the radial nerve substantially increased H-reflex amplitude, yet this result was not observed with a lower stimulation frequency (30 Hz), suggesting a frequency-dependent change in spinal excitability after NMES. These results were recently confirmed with wide-pulse NMES applied on the tibial nerve, with 20-Hz and 100-Hz trains leading to decreased and increased H-reflex amplitude, respectively.^{38,39} However, it should be acknowledged that CNS activation has been shown to differ between wide-pulse NMES applied over the muscle belly and the nerve trunk,^{63,64} making such comparison difficult.

4.4 | Limitations

Some limitations should be acknowledged for the present study. First, five subjects were limited by the maximal stimulator output (100 mA), as they attained this current intensity between the 21st and 25th NMES-evoked contractions. The impact on our results could however be considered negligible, as current intensity remained constant from the 25th NMES-evoked contraction to the end of the NMES protocol to maintain a constant setup for CNS excitability assessments. Second, the number of MVC and evoked responses might have been insufficient to ensure reliability.⁷⁵ However, a low number of MVCs was often used in this type of studies, especially after NMES to avoid recovery.⁵⁴ It was also recently demonstrated that five stimuli produce reliable MEP recordings within one session and our time constraints (~20 s) prevented additional trials to be performed.⁷⁶ Third, the lack of change in evoked EMG responses may be due to the choice to evoke them during contraction, especially with contractions targeting a constant EMG level. As this procedure attempts to hold motoneuron output constant, it is likely to minimize changes in CNS excitability. However, this procedure was necessary to minimize stimulation intensity and the associated discomfort, especially for thoracic stimulation. Fourth, there are differences between SR and H reflex responses. Transcutaneous spinal cord stimulation evokes SR in several muscles and both sides that

may influence each other. Facilitation from close synergists, disynaptic inhibition from antagonists, or influence from the contralateral side could then affect SR amplitude,^{46,77} making interpretation of the change in SR challenging. Finally, changes in CNS excitability induced by wide-pulse NMES have been regularly investigated in a subpopulation of so-called “responders”.^{38,39} A frequency-dependent modulation of spinal excitability was reported in “responders”, whereas unchanged H-reflex amplitude was observed in “nonresponders”.³⁸ This may explain the variability observed in this study, which led to a lack of significant changes in CNS excitability despite a concomitant increase of 16% and 14% of SR and TMEP after wide-pulse NMES at 100 Hz. However, it was not possible to distinguish “responders” and “nonresponders” in the present study, as this classification is often based on the observation of extra torque, which requires a constant and low-current intensity.⁷⁸

4.5 | Perspective

The acute effects of the three tested NMES protocols on neuromuscular function were comparable. Regardless of pulse duration and stimulation frequency, NMES-evoked torque and objective fatigability were similar, and CNS excitability was unchanged after 20 min of quadriceps NMES. Therefore, based on these results, the use of quadriceps NMES at the maximal tolerable current intensity (such as done in clinical practice) with either narrow- or wide-pulses may supposedly lead to comparable treatment effectiveness. Yet, it remains to be demonstrated in healthy and especially clinical populations. For instance, CNS excitability changes should be more expected in patients with neurological conditions, such as stroke or spinal cord injury.^{68,69}

AUTHOR CONTRIBUTIONS

L.E., J.G., G.Y.M., V.R., N.A.M., and T.L. conceived and designed research; L.E., E.L. and C.G.B. performed experiments; L.E. analyzed data; L.E., J.G., G.Y.M., V.R., N.A.M., and T.L. interpreted results of experiments; L.E. prepared figures; L.E., C.G.B., J.G., G.Y.M., V.R., N.A.M., and T.L. drafted manuscript; L.E., E.L., C.G.B., J.G., G.Y.M., V.R., N.A.M., and T.L. approved the final version of manuscript.

ACKNOWLEDGEMENTS

The authors thank all the subjects for their participation and the company Mazet Santé for the loan of the stimulator. This work was supported by a Doctoral Research Grant from University Jean Monnet of Saint-Etienne (to L.E.).

CONFLICT OF INTEREST STATEMENT

No conflicts of interest, financial, or otherwise, are declared by the authors.

DATA AVAILABILITY STATEMENT

The datasets generated during and/or analysed during the current study are available from the corresponding author [TL] on reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Espeit L, Luneau E, Brownstein CG, et al. Acute effects of conventional versus wide-pulse neuromuscular electrical stimulation on quadriceps evoked torque and neuromuscular function. *Scand J Med Sci Sports*. 2023;33:1307-1321. doi:[10.1111/sms.14375](https://doi.org/10.1111/sms.14375)